Receipts OCR and Information Extraction

Computer Vision Project

Group 6

Le Sy Han
Nguyen Quang Minh
Nguyen Cao Manh Thang
Nguyen Trong Bach

Department of ICT
University of Science and Technology of Hanoi

March 2026

Agenda

1. Introduction
2. Dataset
3. Methodology
4. Results
5. Discussion and Future Work
6. Conclusion

Task Description

- Receipt understanding is a document analysis task.
- Input: scanned or photographed receipt image.
- Output: four structured fields:
  - Company
  - Date
  - Address
  - Total

Main challenge

Although the target output is simple, receipt images often contain small text, noisy backgrounds, irregular layouts, blur, shadows, and formatting differences across stores.

Dataset - SROIE2019

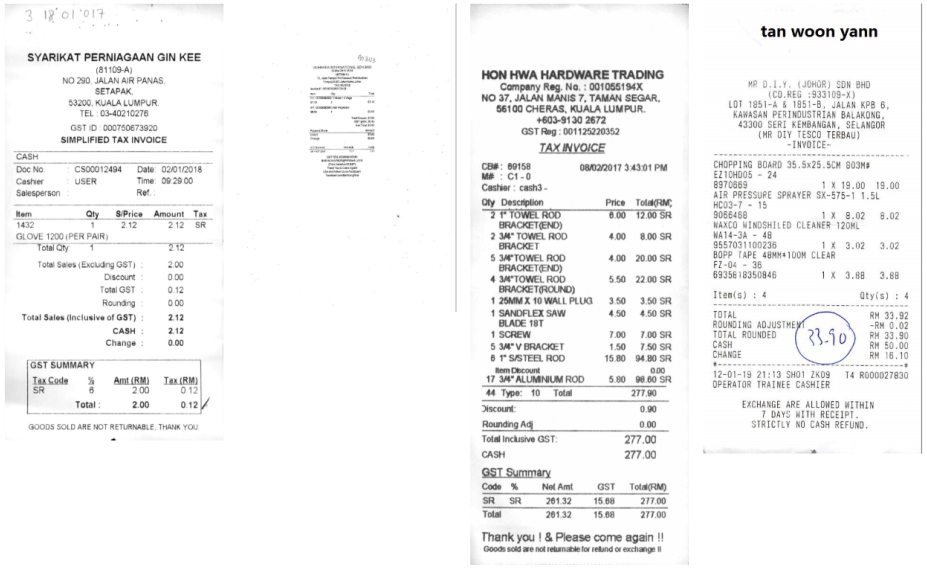


Figure: Sample entries from the Dataset

- Benchmark dataset for receipt OCR and information extraction
- 973 receipt images in total: 626 images for training/validation, 347 images for testing
- 4 Target fields: Company, Date, Address, Total

Methodology Overview

- The current implementation is organized around three main directions:
  - Rule-based system
  - LayoutLMv3-based extraction: Fine-tuned LayoutLMv3 pre-trained model
  - Donut-based end-to-end extraction: Fine-tuned DONUT pre-trained model
- All branches target to extract the same four fields:
  - Company
  - Date
  - Address
  - Total

data_preprocessing.ipynb	Data preparation
ocr.ipynb	OCR support pipeline
rule-based-system.ipynb	EDA for Rule Definition and Rule-based extraction
layoutlm.ipynb	Layout-aware extraction
donut.ipynb	End-to-end extraction

Data Preparation

- Original SROIE files are reorganized into a more convenient structure.
- Each receipt is associated with three aligned files:
  - receipt image,
  - OCR box annotation file,
  - entity annotation file.
- File naming is standardized so later stages can load inputs and labels consistently. (Eg: Image: 000.jpg - Box: 000.txt - Entity: 000.txt)

OCR Support

Preprocessing

- Grayscale conversion
- Gaussian blur
- CLAHE contrast enhancement

PaddleOCR output

- Recognized text
- Bounding boxes
- Confidence scores

Post-processing

- Sort OCR lines
- Filter low-confidence outputs
- Clean short or noisy fragments
- Export cleaned OCR results

OCR Support

OCR Support is used for Rule-based extraction and LayoutLMv3- based extraction.

Rule-based System: Overview (1/2)

- The rule-based system is the simplest baseline in the project.
- It relies OCR output and manually defined extraction rules.
- The goal is to provide a straightforward reference point for comparison.

Rule-based System (2/2)

- After OCR text is cleaned, the system applies **hand-crafted rules**.
- Rules are based on simple textual patterns and heuristic layout assumptions.

Examples of rules:

- Detect **date** using common date formats
- Identify **total** using numeric patterns such as `12.50` or `105.90`
- Extract **company name** from the upper part of the receipt

Strength and Limitation

Easy to interpret and implement, but highly sensitive to OCR errors and layout variations.

LayoutLMv3-based Extraction: Overview (1/3)

- We use **LayoutLMv3**, a multi-modal model that jointly learns from:
  - textual content,
  - spatial layout,
  - document image features.
- This is suitable for receipt information extraction because the meaning of a field often depends on both **text and position** (e.g., totals usually appear near the bottom-right).
- The pipeline consists of three main stages:
  - preprocessing and labeling
  - model fine-tuning
  - prediction reconstruction

Branch logic

Prepare OCR words and boxes → assign token-level BIO labels → fine-tune LayoutLMv3 → reconstruct final fields.

LayoutLMv3: Data Preprocessing and Input Construction (2/3)

- **Text normalization:**
  - convert text to lowercase
  - keep only alphanumeric characters
- **Word-level splitting:**
  - OCR text lines are split into words
  - bounding boxes are duplicated for each word
- **Box normalization:**
  - bounding boxes are scaled to $[0, 1000]$
- **BIO tagging scheme:**
  - B- (begin), I- (inside), O (outside)

LayoutLMv3: Fine-tuning and Prediction Reconstruction (3/3)

Fine-tuning setup

- Base model: `microsoft/layoutlmv3-base`
- Dataset split: **90% training / 10% validation**
- Training uses AdamW optimizer with linear learning rate decay.

Prediction reconstruction

- The model predicts a label for each token.
- Tokens with matching BIO tags are aggregated into entity values.
- Invalid BIO transitions are filtered or corrected using simple heuristics.

Final Output

Reconstructed values for the four target fields: **Company, Address, Date, and Total**.

DONUT: Overview (1/3)

- The third branch uses a fine-tuned **DONUT** model for document understanding.
- Unlike traditional pipelines, it **does not require OCR at inference time**.
- The model receives the **receipt image directly** and generates a structured output sequence.
- This makes the DONUT branch the **most self-contained** among the three approaches.

DONUT: Data Preprocessing and Target Construction (2/3)

- Preprocessing is handled by the `DonutProcessor`.
- Images are converted to RGB and resized to 1280×960.
- **Special tokens** are added for each field (e.g., `<s_company>`).
- The target annotation is converted into a **structured sequence**.

Field	**Tagged format**
Company	`<s_company> ... </s_company>`
Date	`<s_date> ... </s_date>`
Address	`<s_address> ... </s_address>`
Total	`<s_total> ... </s_total>`
Full output	`<s_receipt> ... </s_receipt>`

DONUT: Fine-tuning and End-to-End Generation (3/3)

- Model initialized from `naver-clova-ix/donut-base`.
- Fine-tuned using **Seq2SeqTrainer** with mixed precision (FP16).
- The model learns to **generate structured fields directly**.
- The extraction task becomes a **sequence generation problem**.

Key Idea

Learn a direct mapping from **receipt image** $\rightarrow$ **structured information**.

Normalization and Evaluation Metrics

Output normalization

- Convert text to lowercase
- Trim leading/trailing spaces
- Collapse repeated whitespace

Reason

- Avoid penalizing cosmetic differences

Main metrics

- Field Exact Accuracy
- Precision
- Recall
- F1 Score
- Normalized Levenshtein Distance

Evaluation Protocol

1. Load the receipt sample and ground-truth entities
2. Run one extraction branch
3. Normalize both prediction and reference
4. Compare the four target fields using the selected metrics

Results: Field Accuracy

Model	**Company** ↑	**Date** ↑	**Address** ↑	**Total** ↑
LayoutLMv3	0.285	0.135	0.037	0.669
Rule-based	0.542	0.156	0.009	0.418
DONUT	0.775	0.816	0.550	0.703

- Total is the easiest to detect.
- Address remains the most difficult field across models.
- DONUT performs relatively well on all four fields.

Results: Entity Metrics

Model	**Prec.** $\uparrow$	**Recall** $\uparrow$	**F1** $\uparrow$	**Lev. Dist.** $\downarrow$
LayoutLMv3	0.403	0.282	0.331	0.439
Rule-based	0.281	0.281	0.281	0.416
DONUT	0.711	0.711	0.711	0.102

- LayoutLMv3 identifies only a limited portion of the target fields correctly.
- Rule-based extraction is simpler but remains constrained by OCR quality.
- DONUT gives the strongest overall result in this comparison.

Discussion

- Donut gives the strongest result among the current implementations.
- The two branches using PaddleOCR are still weaker than expected.
- This suggests that OCR post-processing and downstream extraction still need improvement.
- Data augmentation may also be important for improving performance.

Future Work

- Retrain and fine-tune the models more thoroughly
- Improve OCR post-processing
- Explore additional fine-tuning strategies
- Extend the system to Vietnamese receipts
- Move beyond four fixed fields toward more practical applications

Conclusion

- The project studies receipt understanding on the SROIE task.
- Three implementation directions are explored:
  - Rule-based system
  - LayoutLMv3
  - Donut
- All branches target the same four fields: company, date, address, and total.
- Among the current results, Donut is the most promising direction.

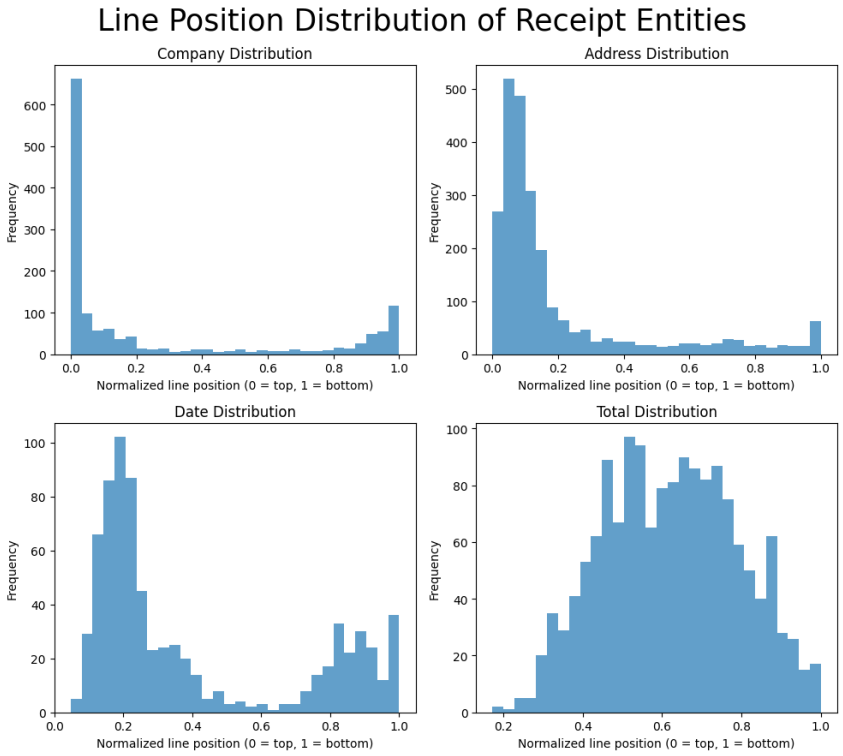
Thank you for listening!

APPENDIX

Exploratory Data Analysis (EDA): Receipt Layout

Goal: Understand spatial patterns of entities in receipts.

- Each OCR text box is assigned a label: `company`, `address`, `date`, `total`, `other`
- Spatial features extracted from bounding boxes:
  - **line_norm**: normalized line order to the range [0,1]
- These features help identify typical regions of entities in receipts.



Layout Observations

- **Company**: top of receipt (line_norm < 0.25)
- **Address**: below company (line_norm < 0.35)
- **Date**: top or bottom
- **Total**: bottom half (line_norm ≥ 0.5)

Exploratory Data Analysis (EDA): Text Patterns

Goal: Identify textual patterns for entity extraction.

Pattern-Based Identification

- **Date patterns**
  - DD/MM/YYYY
  - DD-MM-YYYY
  - DD/MM/YY

  Regex:
  `\d{2}[/-]\d{2}[/-]\d{2,4}`

- **Total amount pattern**
  Example: 12.50, 105.90, 3.80
  Regex: `\d+\.\d{2}`

Frequent Keywords

Address Keywords

Keyword	Freq
no	347
jalan	268
selangor	195
johor	116
lot	107

Company Keywords

Keyword	Freq
bhd	356
sdn	305
gardenia	82
aeon	71
co	45